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Enhancing ESP Efficiency and Reservoir Recovery Through PDG-Based Diagnostics a Gulf of Suez Case Study

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Abstract

This study presents an advanced diagnostic methodology utilizing permanent downhole gauge (PDG) data to evaluate the dynamic behavior and performance of a complex multilayer reservoir in the Gulf of Suez, Egypt, where production is maintained via electrical submersible pumps (ESPs). The primary objective is to enhance reservoir characterization, optimize artificial lift operations, and mitigate operational challenges through the integration of high-resolution pressure and production data. The scope encompasses the assessment of reservoir connectivity, depletion mechanisms, inter-well communication, and post-shutdown production optimization, as well as the evaluation of well intervention efficacy to improve recovery efficiency and operational reliability in a geologically complex environment.

The diagnostic approach leverages high-frequency PDG pressure and production data to perform a detailed reservoir evaluation. Pressure transient analysis (PTA) and material balance techniques are employed to delineate reservoir boundaries and assess connectivity. Inflow performance relationships (IPR) and production data analysis are used to characterize reservoir depletion trends and fluid movement dynamics. Diagnostic fracture injection tests (DFIT) and interference testing are conducted to identify pressure communication between adjacent wells, quantifying the impact of inter-well interference on ESP performance and reservoir depletion. Shut-in pressure buildup analysis is utilized to optimize post-shutdown production strategies, while post-intervention PDG diagnostics are applied to evaluate the effectiveness of well interventions and guide remedial actions.

The analysis yielded significant insights into reservoir dynamics and operational performance. Pressure transient and material balance analyses confirmed reservoir compartmentalization and provided a refined understanding of boundary effects. Interference testing revealed significant pressure communication between wells, leading to suboptimal ESP performance and uneven reservoir depletion. Shut-in pressure buildup analysis facilitated the development of a data-driven well reactivation strategy, achieving a 15% increase in stabilized production rates. Post-intervention diagnostics identified unforeseen wellbore damage, prompting targeted remediation measures, including wellbore cleanout and ESP parameter optimization, which restored production stability and minimized reservoir damage. The study concludes that PDG-driven

diagnostics are indispensable for real-time reservoir surveillance, artificial lift optimization, and sustainable production in complex multilayer reservoirs.

This study introduces an innovative application of PDG data for advanced reservoir diagnostics in ESP-supported wells within a complex multilayer reservoir. The integration of high-frequency pressure and production data enables a comprehensive evaluation of reservoir behavior, inter-well communication, and artificial lift performance. The use of shut-in pressure buildup analysis for production optimization and the application of post-intervention PDG diagnostics for targeted remediation represent significant advancements in reservoir management practices. These methodologies provide a robust framework for addressing operational challenges in geologically complex environments, contributing to the broader field of reservoir engineering and offering actionable insights for similar reservoir systems.

Introduction

Permanent Downhole Gauges (PDGs) have become indispensable tools in modern well surveillance and production optimization strategies. These sensors are typically installed near the reservoir sand-face, in either open-hole or perforated cased-hole completions, where they continuously monitor key downhole parameters such as pressure and temperature throughout the well's production life. The continuous stream of high-frequency data from PDGs enables operators and reservoir engineers to track well behavior, diagnose operational issues, and make informed decisions on real-time optimization of artificial lift systems and reservoir management.

The diagnostic power of PDGs lies in their ability to reflect dynamic reservoir and well conditions, making them valuable for applications such as identifying formation damage, evaluating zonal contributions, understanding the effects of well interventions, and assessing interwell communication and pressure support. These applications are critical for both production and injection wells and are widely documented in the literature ([Shchipanov et al., 2014](#); [Soerawinata and Kelkar, 1999](#); [Kamal et al., 2016](#); [Ribeiro and Horne, 2013](#); [Yousef et al., 2006](#); [Sankaran et al., 2017](#); [Ahmadi et al., 2014](#); [Chorneyko, 2006](#)). In combination with ESP installations, PDG data allow for effective monitoring of pump intake pressure, discharge pressure, temperature variations, and system vibrations. Such monitoring is essential for preventing gas locking, minimizing pump-off conditions, and avoiding premature equipment failure, which in turn extends the ESP run life and improves lift efficiency ([Sankaran et al., 2017](#); [Horne, 2007](#)).

In many modern production environments, Electrical Submersible Pumps (ESPs) are paired with Permanent Downhole Gauges (PDGs) or complete permanent downhole monitoring systems (PDHMS) to deliver continuous measurements of key parameters such as intake pressure, discharge pressure, motor temperature, and vibration. These measurements are critical for protecting the ESP from operational risks such as gas locking, overloading, or underloading, and are essential for optimizing pump performance through dynamic control of Variable Speed Drives (VSDs). In Saudi Aramco fields, for example, PDG-equipped ESP wells form the backbone of intelligent field strategies where production engineers use real-time data for proactive diagnostics and performance tuning ([Mubarak et al., 2010](#); [Gao et al., 2007](#)).

From a reservoir perspective, PDG data are integral to advanced reservoir surveillance techniques such as pressure transient analysis (PTA) and rate-transient analysis (RTA). Continuous measurement allows for identification of reservoir heterogeneities, detection of flow barriers, and recognition of water or gas breakthrough, thereby informing waterflood strategies and recovery planning ([Shchipanov et al., 2014](#); [Soerawinata and Kelkar, 1999](#); [Horne, 2007](#); [Shchipanov et al., 2019](#)). The integration of PDG pressure data with surface flow rates—particularly where high-frequency multiphase flow meters are also employed—further enhances reservoir model calibration and history matching accuracy ([Kamal et al., 2016](#); [Ribeiro and Horne, 2013](#)).

Research by [Athichanagorn et al. \(2002\)](#) and [Liu and Horne \(2012\)](#) emphasized that the long-term acquisition of pressure and temperature data using PDGs enables comprehensive trend analysis, early

detection of pump efficiency losses, and predictive maintenance scheduling. These diagnostic capabilities help extend the mean time between failures (MTBF) of ESP systems and reduce unplanned shutdowns. Neural network-based models, as discussed by [Aminzadeh and Groot \(2000\)](#), have also been applied to interpret PDG data and automatically adjust ESP operating parameters in real time. This form of soft computing reduces reliance on human supervision and improves decision accuracy, especially in complex or offshore settings where timely interventions are challenging.

Despite these advantages, interpreting long-term PDG data remains challenging. The volume of data collected is often enormous and includes measurement noise and outliers, which require careful preprocessing. Moreover, for meaningful reservoir analysis, engineers must isolate specific flow periods such as flowing, shut-in, or transient segments, which represent responses to operational changes. This segmentation is necessary for accurate calculation of reservoir parameters like permeability and skin factor. In ESP-equipped wells, misinterpretation of such data could lead to incorrect assumptions about well performance or reservoir behavior, especially if pressure drops and early transient responses are not clearly distinguished ([Nomura, 2006](#); [Cumming et al., 2014](#); [Liu and Horne, 2012](#)).

Several approaches have been proposed to improve the utility of PDG data for production optimization. Athichanagorn and colleagues proposed workflows that incorporate data denoising, transient segmentation, flow history reconstruction, and behavioral filtering to facilitate the interpretation process ([Athichanagorn et al., 2002](#); [Athichanagorn, 1999](#)). Recent studies have also explored the application of machine learning techniques to automate PDG data interpretation and trend analysis, which is especially useful in high-volume monitoring environments ([Tian and Horne, 2015](#); [Tian and Horne, 2017](#); [Kin, 2001](#); [Tian, 2018](#)).

The synergy between PDG-based diagnostics and reservoir recovery strategies has also been strengthened by the incorporation of smart completions such as inflow control valves (ICVs). Studies by [Ukaegbu and Alyan \(2013\)](#) and [Mubarak et al. \(2010\)](#) demonstrate how combining PDG monitoring with zonal flow control allows operators to adjust production from different laterals based on dynamic reservoir feedback. This selective control is instrumental in delaying water breakthrough, minimizing gas coning, and improving overall recovery efficiency in complex reservoirs. Furthermore, the capability to monitor and remotely control these completions has led to the implementation of closed-loop optimization workflows, where data-driven alerts and optimization algorithms support continuous improvement of well and reservoir performance.

[Al-Yateem et al. \(2014\)](#) presented Saudi Aramco's deployment of intelligent field technologies integrating PDG-equipped ESP wells, real-time monitoring, and smart completions to optimize production and reservoir surveillance. They describe the use of neural networks, SCADA systems, and soft computing for automated well control and enhanced reservoir diagnostics. The study showcases how these technologies improved ESP performance, sweep efficiency, and recovery in both new and retrofitted wells.

This study focuses on the practical implementation of PDG-based diagnostics in wells completed with ESPs in the Gulf of Suez, aiming to improve both artificial lift efficiency and reservoir recovery. By analyzing field data and case histories, the study demonstrates how PDG data can be used to monitor ESP performance, predict failure modes, assess reservoir behavior, and enhance production strategies in a mature offshore field. This integration of real-time downhole monitoring with field-driven optimization provides a valuable framework for extending ESP lifecycle and maximizing oil recovery in similar reservoir settings.

Methods, Procedures, Process

The methodological framework employed in this study is grounded in a comprehensive diagnostic workflow that integrates high-resolution pressure data from Permanent Downhole Gauges (PDGs) with surface production data to analyze the dynamic behavior of a complex multilayer reservoir in the Gulf of Suez, Egypt. This reservoir is produced through Electric Submersible Pumps (ESPs), and the methodology aims

to enhance reservoir understanding, optimize artificial lift operations, and mitigate performance challenges through advanced analytics and real-time diagnostics.

The diagnostic process begins with the application of pressure transient analysis (PTA), which is used to identify reservoir boundaries, heterogeneities, and flow regimes. PTA enables the delineation of reservoir compartments and skin estimation, providing insights into the presence of permeability barriers or conductive faults. To complement this, material balance techniques are employed to assess hydrocarbon-in-place, depletion mechanisms, and the volumetric connectivity between different reservoir layers. Together, these methods establish a structural and dynamic interpretation of the reservoir that guides further diagnostic efforts.

To evaluate fluid inflow behavior and assess production efficiency, Inflow Performance Relationship (IPR) analysis is conducted using both real-time and historical production data. The construction of IPR curves enables the quantification of drawdown performance and identification of potential inflow restrictions or damage. These analyses are further supported by detailed production trend assessments, which help characterize fluid movement dynamics, monitor depletion trends, and flag anomalies in ESP behavior such as load instability or gas locking.

Diagnostic Fracture Injection Tests (DFIT) and interference testing are integral components of the workflow, particularly in assessing inter-well communication. DFITs provide valuable information on fracture closure pressures, leak-off behavior, and formation stress profiles, which are essential for evaluating natural fracture systems and pressure diffusivity. Interference testing, on the other hand, involves monitoring pressure responses in offset wells when one well undergoes rate or shut-in changes. This allows for the detection of hydraulic communication between wells and quantifies the extent of inter-well interference, which can significantly impact the performance and efficiency of ESP systems in closely spaced multilayer reservoirs.

Shut-in pressure buildup analysis is another crucial element, enabling post-shutdown evaluation of reservoir and near-wellbore conditions. By analyzing pressure recovery behavior during shut-in periods, the methodology allows for accurate estimation of average reservoir pressure and skin, helping to refine reservoir models and inform optimal well reactivation strategies. This approach directly supports the design of more effective restart procedures for ESP-driven wells, minimizing risks associated with transient conditions.

Following well interventions, post-intervention diagnostics using PDG data are conducted to evaluate the effectiveness of remedial actions. This includes assessing changes in pressure behavior, drawdown efficiency, and ESP performance in response to interventions such as acid stimulation, mechanical repairs, or wellbore cleanouts. High-frequency PDG data offer immediate feedback on the success of the intervention and guide the need for additional or alternative remedial strategies. This feedback loop is critical for maintaining production stability and ensuring long-term well integrity and ESP reliability.

Overall, the integrated diagnostic methodology presented in this study offers a robust and adaptable approach to reservoir surveillance, artificial lift optimization, and operational troubleshooting. By leveraging the diagnostic power of PDGs in conjunction with classical and modern reservoir engineering techniques, this workflow enables precise identification of production bottlenecks, informed decision-making, and enhanced reservoir management in complex geological settings.

Description of Data Obtained From Different Wells

The six wells analyzed in this study (A-3, A-4, A-5, A-6, A-7, and A-8) are equipped with Electric Submersible Pumps (ESPs) and Permanent Downhole Gauges (PDGs) to support continuous monitoring of downhole and surface operating parameters. Each well is located within a geologically complex multilayer reservoir in the Gulf of Suez and contributes to the field's overall oil production strategy. Detailed datasets were acquired from each well, including high-frequency measurements of intake and discharge

pressures, downhole temperatures, electrical load (current and frequency), wellhead pressure, gross and net production rates, water cut (BS&W), and differential pressure across the pump. These data were analyzed to assess artificial lift performance, identify production trends, and compare reservoir responses under ESP-driven flow conditions. The collective analysis provides a basis for evaluating well health, optimizing lift efficiency, and planning future reservoir management interventions.

The performance of wells A-3, A-4, A-5, A-6, A-7, and A-8 reflects a spectrum of reservoir behaviors and ESP responses, offering valuable insights into the interplay between pressure support, production efficiency, and fluid composition. Intake pressures across the wells vary significantly, with A-3 and A-6 maintaining higher averages above 2200 psi, indicating strong reservoir support, while A-4 operates at a much lower intake pressure of around 636 psi, yet still delivers efficient oil production—demonstrating that productivity is not solely pressure-dependent but also influenced by completion design and fluid mobility.

Discharge pressures remain consistently high across all wells, generally in the range of 2350 to 2800 psi, confirming effective ESP lift and stable hydraulic performance. The corresponding pressure differentials (dp) highlight the strength of artificial lift; A-7 shows the highest differential, exceeding 1000 psi, while A-6 reports the lowest, suggesting possible operational constraints or reduced pump efficiency. Temperature profiles are also consistent, with motor temperatures (Tm) typically ranging from 225°F to 250°F and intake temperatures (Ti) around 185°F to 190°F, indicating normal downhole thermal conditions and reliable motor insulation across the wells.

Electrical indicators such as current and frequency remain within operational norms, with A-5 and A-6 drawing approximately 24–27 amps and A-7 and A-8 slightly higher, nearing 40 amps. Frequency settings vary, with A-5 operating at lower frequencies (~42 Hz), while others, like A-3 and A-7, operate at or above 46 Hz, reflecting tuning strategies adapted to individual reservoir responses. Wellhead pressures (WHP) range from 348 to 608 psi, with lower WHPs generally aligning with higher drawdowns and production rates.

In terms of fluid production, A-5 and A-8 are clearly the highest-yielding wells, each producing over 4000 and 2600 BOPD respectively, with A-8 delivering nearly pure oil flow (BS&W $\approx$ 0.4%) and A-5 following closely (BS&W $\approx$ 10.9%). In contrast, A-6, while maintaining a high gross BFPD, shows clear signs of water encroachment, as its net oil rate is significantly lower—suggesting reservoir heterogeneity or water breakthrough. A-3, A-4, and A-7 display balanced performance, with A-4 notable for its exceptionally low water cut and A-7 achieving a favorable balance of pressure differential and oil rate (~1774 BOPD with ~19% water cut).

The comparative insights gained from analyzing the key performance metrics across these six wells provide a foundation for understanding field-wide production behavior and ESP system dynamics. To further contextualize these observations, a detailed evaluation of each individual well is presented in the following sections. This includes an in-depth review of time-series trends, operational anomalies, and reservoir responses specific to each well, offering a comprehensive diagnostic profile that supports targeted optimization strategies and well-specific intervention planning.

The Actual Data for Well A-4

The time-series dataset acquired from Well A-4 provides a comprehensive view of both surface and downhole operational parameters over an extended monitoring period. Intake pressures (Pi) average around 2540 psi, while discharge pressures (Pd) are higher, averaging 3120 psi, indicating stable ESP operation with an average pressure differential of approximately 571 psi. The motor temperature remains consistent, averaging 233°F, with intake temperatures slightly lower at 191°F, suggesting efficient thermal performance. Electrical characteristics show average current at 42.3 amps and frequency at 41.7 Hz, reflecting stable power delivery to the pump. Wellhead pressure (WHP) averages 545 psi, supporting steady flow conditions at the surface. Despite limited data points for production parameters, the available

records show an average net oil rate of 3155 BOPD and a BS&W content averaging 15.4%, indicating moderate water production. The I2 load metric, with an average of 83.9, further confirms consistent pump performance under varying operating conditions. These measurements collectively provide a robust dataset for evaluating ESP efficiency, identifying operational trends, and supporting diagnostics for reservoir behavior in a multilayer setting.

The Actual Data for Well A-3

Based on the extracted dataset, **Well A-3** demonstrates stable electric submersible pump (ESP) performance with key operational and reservoir parameters consistently tracked over an extended period. The average intake pressure (Pi) is approximately 2227 psi, while the average discharge pressure (Pd) is around 2795 psi, indicating a healthy pressure differential supporting efficient lift performance. The intake and motor temperatures show expected separation, averaging 190°F and 250°F respectively, which is typical for downhole ESP installations and implies effective thermal management. Electrical indicators such as current and frequency further affirm stability, with an average current draw of 23.8 amps and frequency centered around 53.6 Hz.

Surface pressure conditions remain within operational norms, with wellhead pressure averaging 463 psi. Although gross fluid production and oil rates are only partially populated, the available records suggest a gross BFPD of approximately 3120 barrels per day with a net oil production averaging around 2531 BOPD. Water cut, based on the average BS&W content of 23.7%, reflects moderate water handling in the system, common in mature reservoirs. The pump differential pressure (dp) shows an average of 562 psi, with an I2 value (an ESP load indicator) averaging 46.8, suggesting a balanced hydraulic and electrical load on the pump. Together, these parameters highlight a well-performing ESP system with a steady flow regime, offering reliable data for deeper reservoir diagnostics and artificial lift optimization.

The Actual Data for Well A-5

Well A-5 demonstrates robust ESP operation and high production efficiency. The average intake pressure (Pi) is approximately 2132 psi, while the discharge pressure (Pd) is around 2733 psi, yielding a stable pressure differential of nearly 600 psi, which supports efficient lifting. Intake and motor temperatures average 187°F and 232°F, respectively, reflecting healthy downhole thermal conditions. The current draw is consistently around 24.3 amps, paired with a frequency near 42 Hz, confirming that the electrical drive system is operating near its design load range.

Surface pressure measurements (WHP) average about 608 psi. Although production data was partially recorded, the available entries indicate a high gross fluid rate averaging 4546 BFPD and net oil production of around 4194 BOPD. The relatively low average water cut (BS&W $\approx$ 10.9%) confirms the well is yielding high-quality crude with minimal water handling requirements. The load indicator (I2) averages around 48, showing balanced performance with no signs of overloading. Some data anomalies (such as minimum Pi and Pd values close to 300 psi and zero-frequency readings) suggest brief shut-ins or sensor dropouts, but they are rare and do not distort the overall trend.

The Actual Data for Well A-6

Well A-6 demonstrates high-frequency surveillance with consistent electric submersible pump (ESP) performance across more than 2,100 recorded entries. The average intake pressure (Pi) of approximately 2201 psi and discharge pressure (Pd) of 2801 psi indicate a stable pressure differential that supports efficient reservoir lifting. Temperature readings show an average intake temperature (Ti) of 189°F and a motor temperature (Tm) of 202°F, which are within safe operating ranges and reflect controlled thermal loading on the ESP motor.

The electrical current remains highly stable with a narrow standard deviation, averaging 27.4 amps, while frequency readings are steady at 40 Hz, implying a stable drive system. Wellhead pressure averages around 410 psi. Though production data is more limited in frequency, gross production rates where available suggest an average of 3060 BFPD with a corresponding average net oil production around 600 BOPD. The high water cut inferred from the limited BS&W entries indicates a mature reservoir under waterflood or natural water drive, contributing to the large fluid volume.

The ESP load indicator (I2) averages around 54.7, and the pump differential pressure is consistently recorded at 39 psi, further affirming operational consistency. A few anomalous entries in net oil production (negative values) likely reflect data acquisition noise or temporary flow disturbances, which can be filtered in post-processing. Overall, A-6 appears to be a high-volume fluid producer with steady ESP dynamics, albeit with declining oil cut, making it a candidate for water handling optimization and potential zonal isolation review.

The Actual Data for Well A-7

Well A-7 exhibits reliable ESP operation with high-frequency data monitoring across more than 2700 entries. The average intake pressure (P_i) is approximately 1758 psi, while the average discharge pressure (P_d) is 2812 psi, yielding a strong pressure differential of over 1000 psi that supports robust artificial lift efficiency. The downhole temperatures are well-regulated, with average intake and motor temperatures of 189°F and 225°F, respectively, indicating stable thermal loading on the motor and surrounding environment.

Electrical parameters remain tightly controlled, with average current at 40.3 amps and frequency around 46 Hz, confirming consistent power delivery and motor control. The wellhead pressure averages around 348 psi, with operational peaks supporting higher drawdowns. Although full production data is only available for a subset of the records, these show an impressive gross fluid rate averaging 2096 BFPD and a net oil rate of approximately 1774 BOPD. This translates to a favorable water cut of about 19%, positioning the well as a high-yielding oil producer with moderate water handling needs.

The pump differential pressure (dp) averages 1056 psi, which corroborates the strong lifting performance observed in the pressure differential data. The I2 indicator is not recorded, but the high volume and thermal-electric balance suggest the ESP system is operating well within its designed capacity. Some minor anomalies in the dp column (including values of -1) may reflect brief shutdowns, system resets, or sensor noise but do not significantly affect the interpretation. Overall, A-7 is among the most productive wells in the field, contributing substantially to total field output with a well-balanced ESP profile.

The Actual Data for Well A-8

Well A-8 shows high stability and sustained productivity supported by robust ESP operation. The average intake pressure (P_i) is approximately 1811 psi, while the discharge pressure (P_d) averages 2574 psi, yielding a consistent differential pressure (dp) of around 758 psi. This differential confirms healthy lift conditions, with only minor deviations observed. Temperature readings at the pump intake (T_i) and motor (T_m) average 186.5°F and 235.3°F respectively, indicating well-regulated thermal conditions within the downhole assembly.

The ESP motor operates consistently at a fixed frequency of 40 Hz, drawing an average current of 35.5 amps with minimal variation. This stability implies that the pump is functioning near its designed load range. Wellhead pressure (WHP) readings average 430 psi, aligning with the strong downhole drawdown performance.

Production data, although recorded for a smaller subset of the dataset, confirms high-efficiency oil production. Gross fluid production rates average 2619 BFPD, with an exceptionally low water cut (BS&W $\approx 0.43\%$), resulting in a net oil output of 2609 BOPD—nearly 100% of the gross flow. The load indicator (I2) averages 69.7, showing efficient energy transfer and absence of mechanical overloads. Occasional

anomalies such as dp values as low as -10 psi likely represent data noise or brief shutdown events, but they are statistically insignificant given the sample size.

Application of The Approach on Data

From the continuous datasets recorded by the downhole gauges installed in the studied wells, several operational anomalies and reservoir responses were identified and investigated individually. Each case was analyzed as a separate diagnostic scenario to uncover the underlying causes of performance deviations, with the aim of either optimizing production or resolving persistent issues. This targeted approach allowed for the integration of high-frequency pressure and temperature data with operational knowledge to draw actionable conclusions. In this study, we present three representative cases that illustrate this methodology. The first case focuses on understanding inter-well connectivity by identifying simultaneous and abrupt pressure changes recorded in two different wells, suggesting pressure communication between them. This behavior was detected through detailed trend analysis, which revealed aligned pressure drops across the two wells, indicating potential reservoir continuity or interference.

Case One: Diagnosing Interwell Connectivity and Reservoir Depletion Behavior

This case highlights the detection of interwell pressure communication and the evaluation of localized depletion behavior within the studied reservoir using high-resolution PDG data. The analysis began with plotting the Rate Normalized Pressure (RNP) against material balance time on a log-log scale for multiple wells, as shown in Figure 1 (*Reservoir Drainage Behavior and Depletion Diagnostics*). The RNP plots revealed distinct reservoir responses: Well A-5 exhibited the steepest depletion trend, indicating early transition into boundary-dominated flow (BDF) and suggesting that this well, although exhibiting the highest productivity index (kh), is draining from a more confined and possibly bounded zone. This accelerated depletion was likely exacerbated by competitive drawdown from nearby wells.

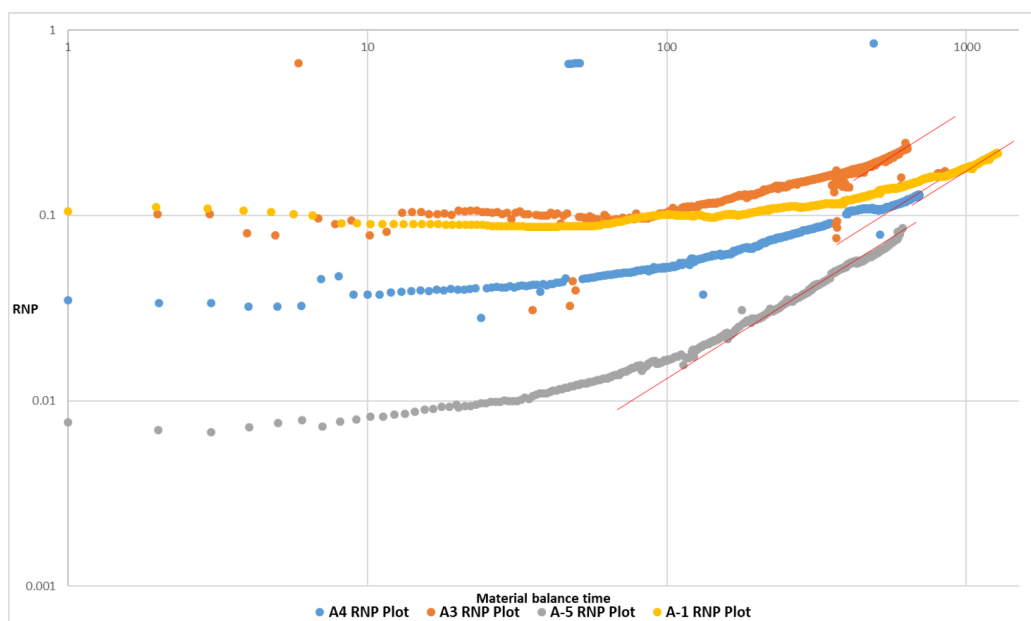


Figure 1—Rate Normalized Pressure (RNP) against material balance time on a log-log scale for multiple wells

In contrast, Well A-1 demonstrated a much flatter RNP trend over an extended period, implying minimal depletion and efficient drainage from an isolated reservoir compartment. The flat RNP curve suggests

sustained support from a less-depleted eastern portion of the field, positioning this area as a promising candidate for infill drilling. These drainage profiles provided critical input for refining the dynamic reservoir model and identifying zones of reservoir heterogeneity.

Further evidence of interwell connectivity was established through time-synchronized pressure responses in Wells A-4 and A-5. On July 13, 2023, a sudden deviation in intake pressure (PIP) and differential pressure (ΔP) was recorded simultaneously in both wells (Figures 2 and 3), despite the absence of any operational changes in ESP settings. This synchronized pressure anomaly is a hallmark indicator of interference between the two wells and confirms direct pressure communication across their drainage areas. By correlating the timing of the pressure trend deviation in both wells, the reservoir team validated the presence of shared connectivity pathways, an insight that was previously uncertain based on static geological models alone.

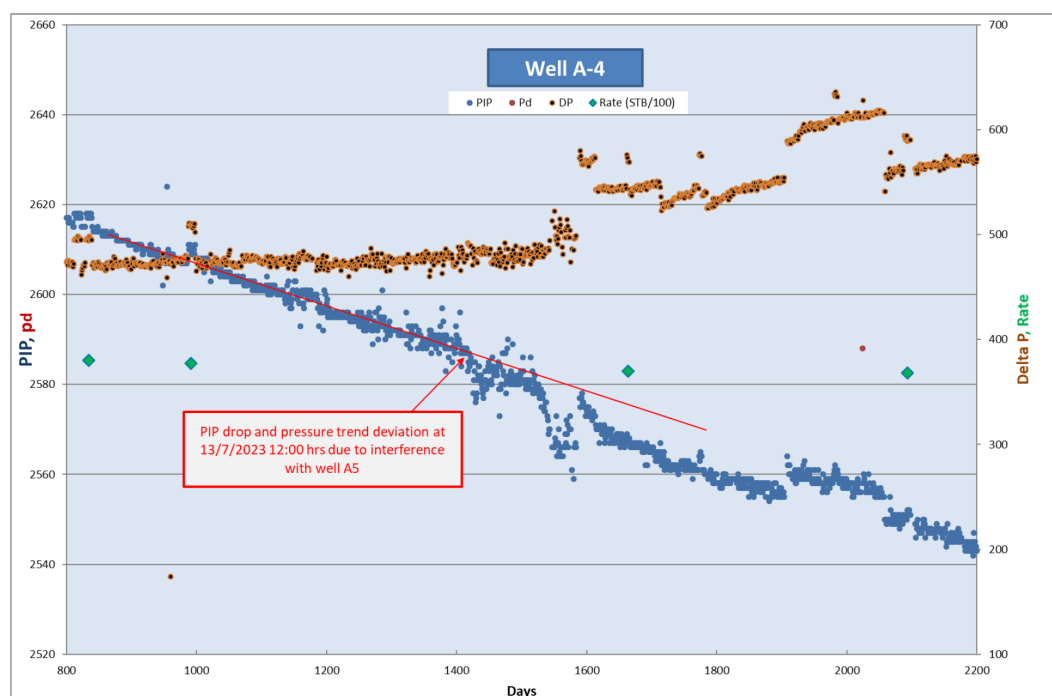


Figure 2—intake pressure (PIP) and differential pressure (ΔP) well A-4

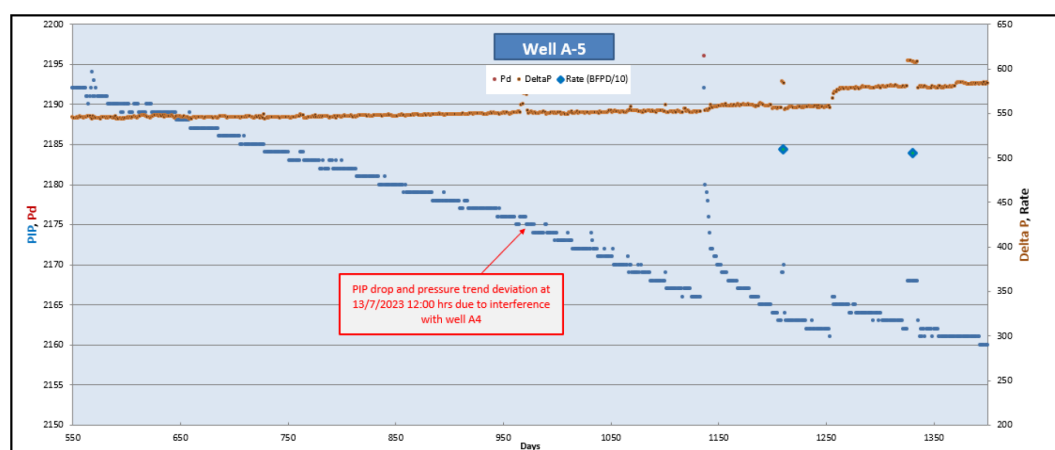


Figure 3—intake pressure (PIP) and differential pressure (ΔP) well A-5

The implications of this diagnostic event were significant. Firstly, it confirmed dynamic reservoir continuity between A-4 and A-5, reducing geological uncertainty and allowing for improved calibration

of the reservoir simulation model. Secondly, it provided a basis for optimizing well spacing to mitigate interference and improve pressure support strategies. Lastly, it enabled the team to update history-matching assumptions with real field data, resulting in more accurate forecasts for recovery and development planning.

This case illustrates how PDG-based surveillance, when coupled with dynamic trend analysis, can expose interwell interactions and guide spatial reservoir characterization. The ability to detect such interference without requiring operational disturbances or flow tests demonstrates the power of continuous downhole monitoring in unlocking the full diagnostic value of the reservoir.

Case Two: Diagnosing Production Bottlenecking and Backpressure Effects

This case focuses on diagnosing production bottlenecks and surface backpressure effects within the platform network, as detected through anomalies in downhole pressure data following the commissioning of a new well. The addition of Well A-8 to production on July 27, 2023, triggered a noticeable and immediate impact on existing wells, particularly A-4 and A-5. As shown in Figures 4 and 5, both wells experienced a sudden and simultaneous increase in pump discharge pressure (Pd) and differential pressure across the ESP (ΔP) at the exact time Well A-8 began producing. This abrupt change is a clear signal of increased backpressure likely induced by constrained surface handling capacity or suboptimal manifold configuration.

The high backpressure resulted in unstable flow conditions and performance degradation. Notably, post-shut-in recovery behavior became inconsistent, subsequent production from A-4 and A-5 exhibited strong dependency on the sequence in which the wells were brought online. When the wells were restarted following shut-ins, those initiated earlier would achieve higher production rates, while those started later suffered from lower output due to residual system pressure. This effect, which became increasingly evident in the latter half of the dataset, further confirmed the presence of a surface facility bottleneck affecting dynamic well performance.

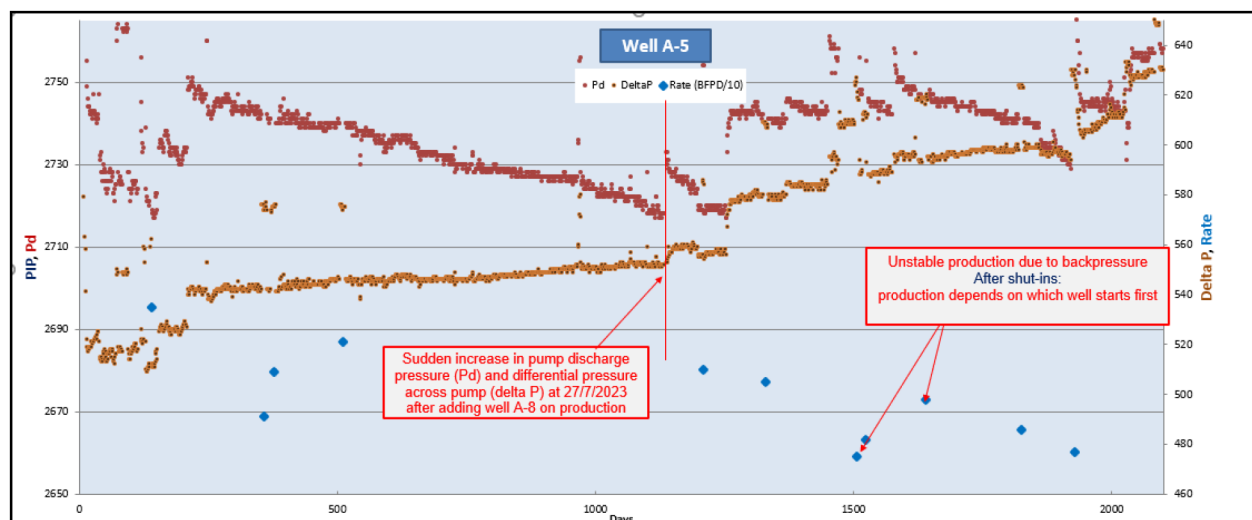


Figure 4—intake pressure (PIP) and differential pressure (ΔP) Well A-5

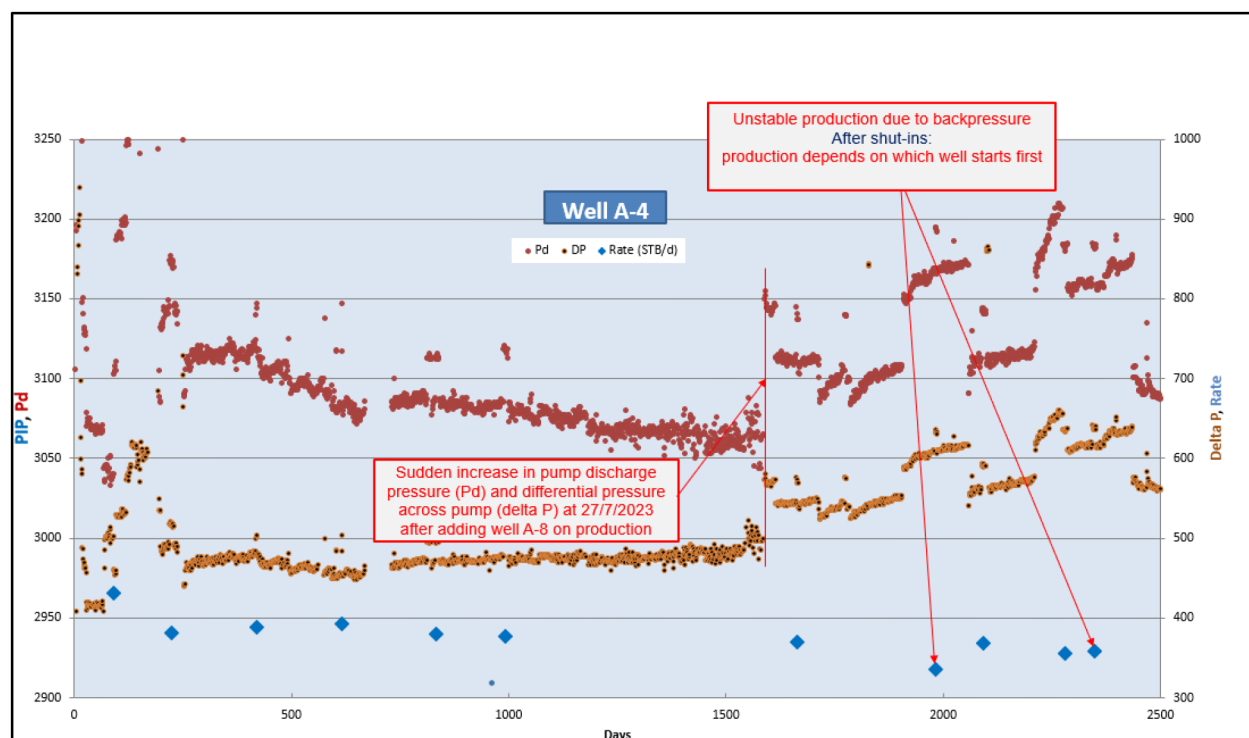


Figure 5—intake pressure (PIP) and differential pressure (ΔP) well A-4

This diagnosis led to a significant shift in development planning. Rather than continuing with the previously envisioned strategy of drilling additional wells from the same platform, operational focus shifted toward optimizing startup sequences and managing surface pressure interactions. In particular, sequencing production based on wells with lower water cuts or higher potential was found to maximize total system output. The analysis demonstrated how real-time PDG monitoring can reveal the complex interplay between subsurface flow and surface constraints, allowing for production optimization without additional drilling investments.

Case Three – Diagnosing Flow Efficiency and Reservoir Depletion Behavior Using RNP and ESP Diagnostic Data

In this case, we investigated long-term reservoir performance and ESP behavior, focusing on the pressure and production diagnostics of well A-4. The analysis integrates data from the ESP pump performance, production metrics, and rate-normalized pressure (RNP) plots to evaluate flow efficiency, depletion trends, and water cut dynamics.

The pump performance chart [Figures 6](#) reveals that after the initial production phase, a noticeable increase in pump discharge pressure (Pd) and differential pressure (ΔP) occurred on 27/3/2023, coinciding with a choke adjustment from 31/64" to 30/64". This change, while minor in mechanical terms, appears to have significantly influenced the fluid flow regime, as confirmed by the concurrent rise in Pd and relatively stable intake pressure (Pi). Meanwhile, the BS&W (Basic Sediment and Water) level shows a steep rise, signaling growing water encroachment or channeling within the reservoir.

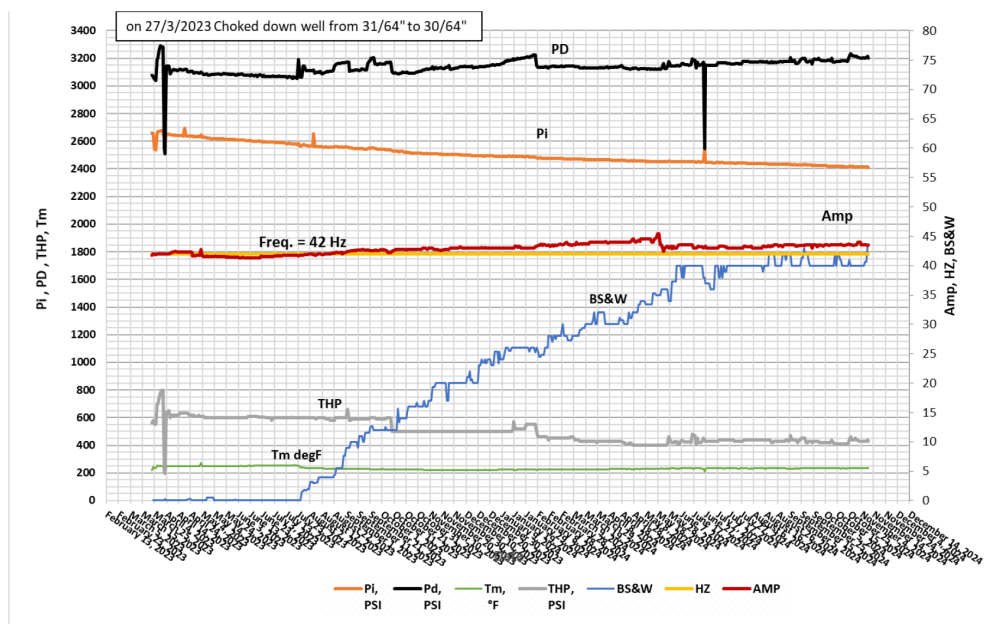


Figure 6—The pump performance chart Well A4

Parallel trends in the production performance plot [Figures 7](#) highlight a steady increase in BS&W from zero to over 40% within a year, alongside a significant drop in oil production rate (BOPD), which suggests progressive water breakthrough and reservoir depletion. Interestingly, despite a decline in oil rates, total fluid production (BFPD) remains relatively stable, implying water production dominance. This evolving water cut significantly affects ESP efficiency and inflow behavior.

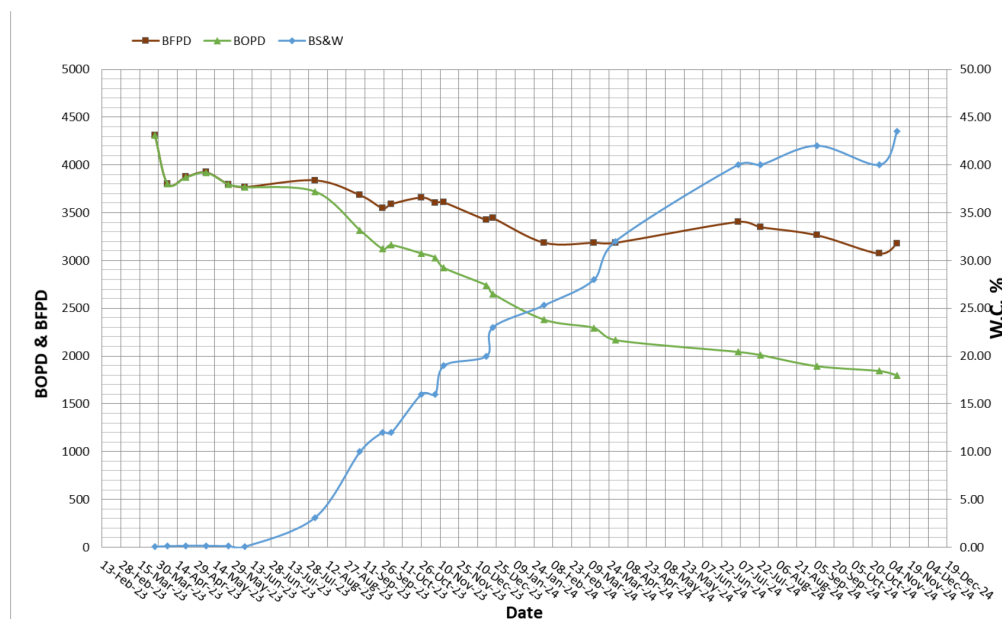


Figure 7—Parallel trends in the production performance plot Well A4

The RNP diagnostic plot [Figures 8](#) further supports this interpretation. Well A5, which exhibits the steepest RNP curve, indicates advanced depletion due to high transmissibility (kh) and likely exposure to no-flow boundaries or competitive production from adjacent wells. In contrast, well A1, situated in the eastern portion of the field, demonstrates a much flatter RNP trend over time, indicating minimal depletion and

high drainage efficiency. These findings imply that the eastern part of the reservoir remains underexploited and is potentially suitable for future infill drilling.

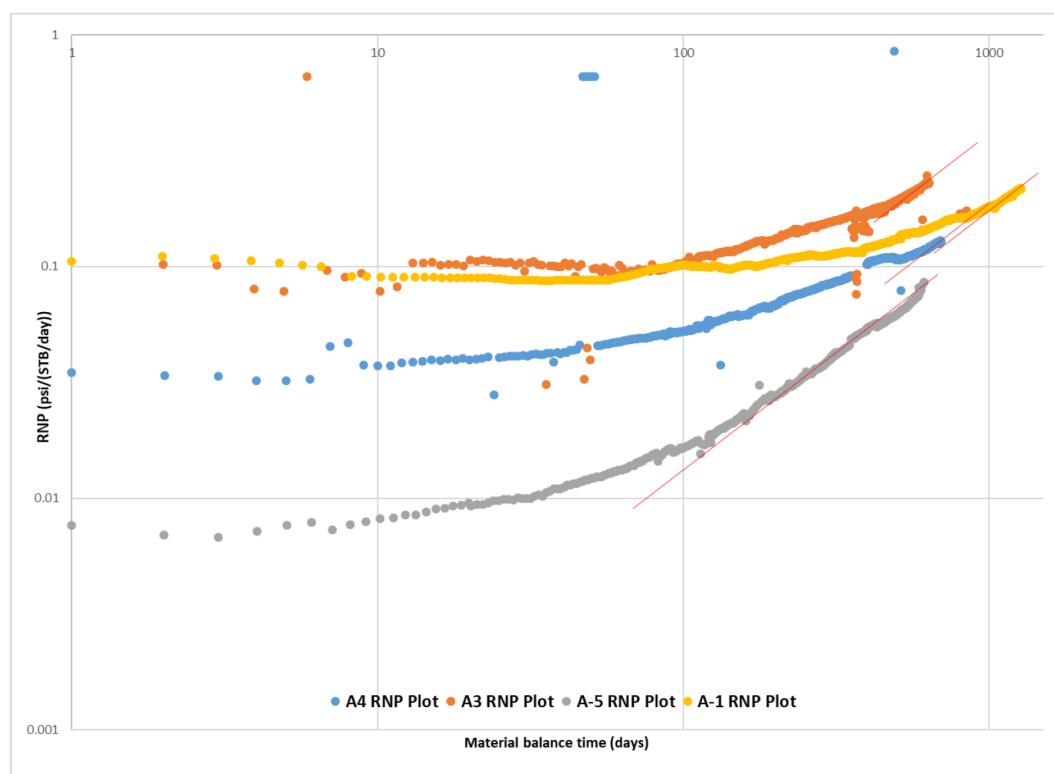


Figure 8—The RNP diagnostic plot

Together, these insights highlight the role of continuous ESP monitoring and RNP diagnostics in understanding not only well performance but also underlying reservoir dynamics. Such integration allows for informed decisions on production optimization, artificial lift management, and field development planning.

Results and Discussion

The application of a PDG-based diagnostic workflow across six wells in a complex multilayer reservoir has demonstrated significant value in enhancing both artificial lift efficiency and reservoir management. By systematically analyzing real-time pressure and production data, the study revealed clear patterns of reservoir behavior, interwell interaction, and ESP response that would have been difficult to capture using conventional surveillance methods alone.

One of the most notable outcomes of this integrated approach is the ability to correlate subtle changes in downhole pressure signatures with broader operational events and reservoir dynamics. The continuous and high-resolution nature of PDG data allowed for the early identification of performance anomalies, pressure trends, and depletion signatures, enabling faster and more informed decisions. The diagnostic framework proved particularly effective in isolating causes of suboptimal performance, such as backpressure effects and reservoir interference, while also highlighting opportunities for production optimization through operational re-sequencing and equipment tuning.

Furthermore, the integration of rate-normalized pressure (RNP) analysis added an important dimension to the interpretation of well performance over time. This provided a consistent method to assess reservoir support, drainage efficiency, and depletion behavior, ultimately contributing to better reservoir segmentation

and more strategic development planning. Importantly, the approach enabled proactive field management by aligning well-level behavior with platform-wide constraints and recovery goals.

Overall, the results affirm that PDG-based diagnostics, when applied through a structured analytical process, can transform well surveillance from a reactive to a predictive and optimization-driven practice. The insights generated from this workflow offer a scalable methodology for real-time reservoir evaluation, ESP troubleshooting, and long-term performance forecasting in complex production environments.

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